Evaluating the Impact of Narrative-Based Augmented Reality on Early STEAM Learning Outcomes in Sri Lankan Primary Education

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Primary classrooms often consist of learners with different levels of understanding, making it challenging for traditional teaching methods to effectively engage all students. Measurement concepts such as length, mass, time, capacity, and data representation are typically taught using static textbook diagrams, which may limit conceptual understanding for young learners. This research proposes the development of a marker-based AR measurement storybook designed for primary school students. The system integrates narrative storytelling with interactive AR scenes that visualize measurement tools such as rulers, weighing scales, stopwatches, measuring cylinders, and bar charts. Developed using Unity and Vuforia, the application allows students to scan printed storybook pages to activate corresponding 3D learning scenes on a mobile device. The goal of the system is to enhance engagement and improve conceptual understanding through visual and interactive learning experiences. User evaluation will be conducted with primary school students to assess the effectiveness of the AR learning environment in supporting measurement education.

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LIST OF ABBREVIATIONS

Abbreviation	**Definition**
AI	Artificial Intelligence
STEAM	Science, Technology, Engineering, Art and Mathematics
AR	Augmented Reality
3D	Three Dimensional
SDK	Software Development Kit

1. INTRODUCTION

1.1 Background and Context

STEAM education plays a critical role in developing analytical thinking, problem-solving abilities, and scientific literacy among students. Early exposure to STEAM concepts during primary education is essential because the knowledge and skills developed at this stage form the foundation for future academic achievement. However, many primary school students struggle to understand STEAM concepts when they are presented only through textbooks and verbal explanations.

In the Sri Lankan primary education system, Grade 4 and 5 students often face difficulties in understanding measurement concepts in mathematics, including length, mass, capacity, time, and data interpretation. These topics require students to visualize quantities and understand how measurement tools function in real-world contexts. However, traditional classroom teaching mainly relies on static textbook diagrams and teacher explanations, which limit students' ability to develop a clear conceptual understanding.

Globally, similar challenges exist in STEAM education, particularly in topics that require spatial and process-based understanding. AR has emerged as a promising educational technology capable of addressing these challenges by overlaying interactive digital content onto the real world. AR allows learners to visualize abstract concepts through 3D models and interactive simulations, which improves engagement and conceptual understanding [1], [2]. Research has also shown that AR can enhance learning motivation and support experiential learning environments in primary education [3], [4].

Another emerging approach in educational technology is the integration of storytelling with interactive learning tools. Narrative-based learning environments provide meaningful contexts that help students connect educational concepts with real world situations. AR storybooks combine narrative storytelling with interactive visualizations, enabling students to explore educational concepts through immersive experiences [5].

Despite the global growth of AR based learning systems, there are currently no AR learning tools specifically designed for the Sri Lankan primary mathematics curriculum, particularly for measurement topics. Most existing AR educational applications are developed for different curricula or for secondary and higher education levels. As a result, Sri Lankan primary school students have limited access to interactive digital learning environments that support conceptual understanding of measurement.

To address this gap, this research proposes a narrative-based mobile AR measurement storybook designed for Grade 4 and 5 students. The system integrates a printed storybook with mobile AR application. When students scan each storybook page using a mobile device, an interactive AR scene appears, allowing them to explore measurement tools such as rulers, weighing scales, stopwatches, and measuring cylinders in real time.

The proposed solution also aligns with Sustainable Development Goal (SDG) 4 – Quality Education, which emphasizes the importance of inclusive and high-quality learning opportunities through innovative educational technologies.

1.2 Literature Review

AR has been widely studied as a tool to improve STEAM learning by making abstract concepts interactive and visually accessible. Systematic reviews indicate that AR enhances conceptual understanding, engagement, and motivation in primary and secondary education [1], [2], [4]. AR also supports digital literacy and experiential learning, allowing students to explore concepts actively rather than passively [3], [10].

Narrative based AR storybooks have been explored as a method to context-based learning. Faria [6] showed that storytelling in AR helps students connect abstract concepts to real world scenarios, while Wang et al. [5] introduced Metabook, a system for automatically generating interactive AR storybooks. Other studies demonstrate that AR improves achievement in STEAM subjects [7], [8], [9], but most focus on secondary education or early childhood learning, with little attention to primary grade mathematics.

Despite these benefits, current AR systems exhibit several limitations. Most applications are designed for Western curricula and secondary learners, limiting relevance for South Asian primary students [4], [6]. Existing AR storybooks often lack interactivity or focus on passive visualizations rather than hands-on concept manipulation [3], [9]. Importantly, few systems provide bilingual content or integrate a narrative framework aligned with local contexts, which is crucial for meaningful learning in Sri Lankan classrooms [4], [8]. Furthermore, measurement topics in primary mathematics such as length, mass, capacity, and data interpretation are largely unaddressed in existing AR tools [1], [2].

1.3 Research Gap

The main research question of this study is whether a narrative based mobile AR system can improve STEAM concept understanding and knowledge retention in Sri Lankan Grade 4-5 students compared to conventional textbook based instruction. While AR has been widely recognized as a tool for enhancing engagement, comprehension, and retention in STEAM education [1]– [3], current AR applications largely focus on secondary or early childhood education and are rarely adapted for primary-grade learners [1], [2]. Most existing systems are developed for Western curricula, limiting their cultural relevance, and do not address the specific challenges faced by Sri Lankan students in topics such as measurement, where abstract concepts require real-world visualization [4], [6], [8].

Current teaching methods such as textbooks, static diagrams, and teacher explanations fail to provide interactive, experiential learning necessary for students to understand units of length, mass, capacity, time, and data representation. Though some AR storybooks exist, they often provide passive visualizations rather than allowing students to manipulate virtual measurement tools, and they rarely integrate a narrative that creates meaningful context for learning [5], [6], [9]. Furthermore, bilingual content is almost entirely absent in existing AR systems, which limits accessibility in multilingual settings like Sri Lanka [4], [8].

This approach is novel because it combines interactive AR simulations with a narrative framework, culturally grounded in a Sri Lankan context, and offers bilingual support in Sinhala and English. The proposed system allows students to actively engage with measurement instruments, observe results in real time, and connect abstract mathematical concepts to meaningful real-world scenarios. Evidence from prior studies demonstrates that AR improves conceptual understanding and retention [1]– [3], while narrative-based learning contextualizes abstract knowledge effectively [5], [6]. By integrating these elements specifically for Sri Lankan Grade 4–5 measurement topics, this study addresses a clear gap in both local and global educational research, providing a standalone, innovative, and culturally relevant tool that current approaches do not offer.

Main Research Question:

Can a narrative-based mobile AR system improve STEAM concept understanding and knowledge retention in Sri Lankan Grade 4–5 students across mathematics and science topics compared to conventional textbook-based instruction?

Sub-Research Questions:

- SRQ1: Does a narrative-driven AR measurement experience improve conceptual understanding and practical application accuracy for measurement topics (length, mass, capacity, time, and data) among Sri Lankan Grade 4–5 students compared to conventional textbook-based instruction?
- SRQ2: Does a narrative AR storybook depicting biological life cycles and ecosystem processes improve scientific process understanding and knowledge retention in Sri Lankan Grade 4–5 students compared to static diagram-based textbook instruction?
- SRQ3: Does an interactive AR physical science storybook improve conceptual understanding of light, shadow, and forces and reduce persistent misconceptions in Sri Lankan Grade 4–5 students compared to teacher-led demonstration with static diagrams? (This proposal focuses on SRQ3.)
- SRQ4: Does a narrative AR storybook embedding fraction, decimal, and number operation concepts in a real-world context improve fraction conceptual understanding and number operation accuracy in Sri Lankan Grade 4–5 students compared to conventional symbolic instruction?

2. OBJECTIVE

2.1 Main Objective

To design, develop, and evaluate a set of four narrative-based mobile Augmented Reality storybook applications that improve STEAM concept understanding and knowledge retention among Sri Lankan Grade 4–5 students across mathematics and science topics, where the Art component is expressed through culturally grounded Sri Lankan narrative story arcs and visual aesthetics embedded in the printed storybook design, compared to conventional textbook-based classroom instruction.

2.2 Specific Objectives

1. To develop a narrative-driven AR measurement storybook application that makes length, mass, capacity, time, and data concepts visually concrete for Grade 4–5 mathematics students through interactive 3D measurement instrument simulations. (Component 1 — this proposal)

2. To develop a narrative-driven AR life science storybook application that animates biological processes plant life cycles, food chains, and ecosystems using locally grounded Sri Lankan species to improve scientific understanding in Grade 4–5 Parisaraya students. (Component 2)

3. To develop a narrative-driven AR physical science storybook application that makes light ray propagation, shadow formation, and force dynamics spatially visible and interactively manipulable for Grade 4–5 Parisaraya students, targeting the misconceptions that arise from static, diagram-only instruction. (Component 3)

4. To develop a narrative-driven AR number concepts storybook application that embeds fractions, decimals, place value, and number operations within a real-world bakery context to improve number concept understanding and operational accuracy in Grade 4–5 mathematics students. (Component 4)

3. METHODOLOGY

3.1 Research Approach

The research adopts a Design Science Research methodology, which focuses on creating and evaluating innovative technological artifacts to solve real-world problems. In this study, the artifact is a narrative-based mobile AR learning system that integrates a printed storybook with interactive AR scenes.

To evaluate the effectiveness of the system, an experimental learning study will be conducted with Grade 4–5 students. The evaluation will compare the learning outcomes of students using the AR system with those using conventional textbook-based instruction. This approach allows the study to measure the impact of AR-based learning on conceptual understanding and practical application of measurement concepts.

3.2 Technical Methods and Algorithms

The proposed system consists of two main components: a printed narrative storybook and a mobile AR application. Each page of the storybook contains a visual marker that triggers a corresponding AR scene when scanned using a mobile device.

Each AR scene visualizes a measurement activity connected to the narrative stage of the story. Interactive elements include:

- A 3D ruler that visually measures the length of the running track.
- A virtual weighing scale that displays the mass of sports equipment.
- A 3D stopwatch that allows students to read race timings.
- A measuring cylinder simulation that fills to specific capacity levels.
- A dynamic bar chart that builds itself as race results are recorded.

A Narrative Manager script implemented in Unity controls the sequence of AR interactions for each story page. This script ensures that the correct AR scene appears when a page is scanned and that user interactions are linked to the appropriate learning activity. Immediate visual feedback is provided when students select or read measurement values correctly.

3.3 Tools and Platforms

Table 1

Tool	Purpose	Justification
Unity Game Engine	Development of the mobile AR application and interactive learning scenes.	Unity provides a powerful real-time development environment with strong support for 3D graphics, user interaction, and educational simulations.
Vuforia AR SDK	Image recognition and marker-based augmented reality tracking of storybook pages.	Vuforia offers reliable image-target detection and seamless integration with Unity, making it suitable for implementing marker-based AR systems using printed storybooks.
Blender	Create and optimize 3D models such as rulers, weighing scales, stopwatches, and measuring cylinders.	Blender is a free and open-source 3D modeling tool that allows efficient creation of lightweight 3D assets suitable for mobile AR applications.
Adobe Illustrator / Canva	Design of the printed storybook pages and visual markers used for AR tracking.	Allow high-quality graphic design and layout creation, ensuring that storybook images are visually appealing while maintaining clear markers for accurate AR detection.
Data Logging & Analytics	Statistical analysis of collected data such as pre-test and post-test scores.	Calculate descriptive statistics and compare learning performance to evaluate the effectiveness of the AR learning system.
Mobile Testing Devices	Test the AR learning system in real-world conditions.	The system is designed for smartphone-based interaction with the printed book.

3.4 Data Collection and Ethical Considerations

Data collection will be conducted through an experimental study involving Grade 4–5 students. **Two groups of students will participate:**

- Control Group: Students learn measurement concepts using traditional textbook-based instruction.
- Experimental Group: Students learn using the narrative-based AR storybook system.

Data will be collected through:

- Pre-test and post-test assessments to measure conceptual understanding of measurement topics.
- Observation of student interaction during AR activities to assess engagement.
- Short usability questionnaires to collect feedback on user experience and learning effectiveness.

Ethical considerations are important when conducting research with children. Participation will be voluntary, and parental or institutional consent will be obtained before conducting the study. No personal identifying information will be collected, and all data will be used strictly for research purposes.

3.5 Validation Strategy and Performance Metrics

The effectiveness of the proposed system will be evaluated using several performance indicators.

First, learning improvement will be measured by comparing pre-test and post-test scores between the experimental and control groups. An increase in post-test scores among students using the AR system will indicate improved conceptual understanding.

Second, task accuracy will be measured during AR interactions, such as correctly reading measurement values or identifying appropriate units.

Third, student engagement and usability will be evaluated using questionnaire responses and observational data during the learning sessions.

These metrics will allow the study to determine whether the narrative-based AR measurement system improves learning outcomes and provides a more engaging educational experience compared to traditional teaching methods.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed system is a mobile AR learning platform that integrates a printed narrative storybook with an interactive AR application to support STEAM learning for Grade 4–5 students. The system allows students to scan storybook pages using a mobile device, triggering AR scenes that visualize educational concepts related to measurement.

Each storybook page acts as an Image Target marker, which is recognized by the AR system. Once the marker is detected, the system displays the corresponding AR learning scene aligned with the narrative stage of the story.

The data flow within the system follows these steps:

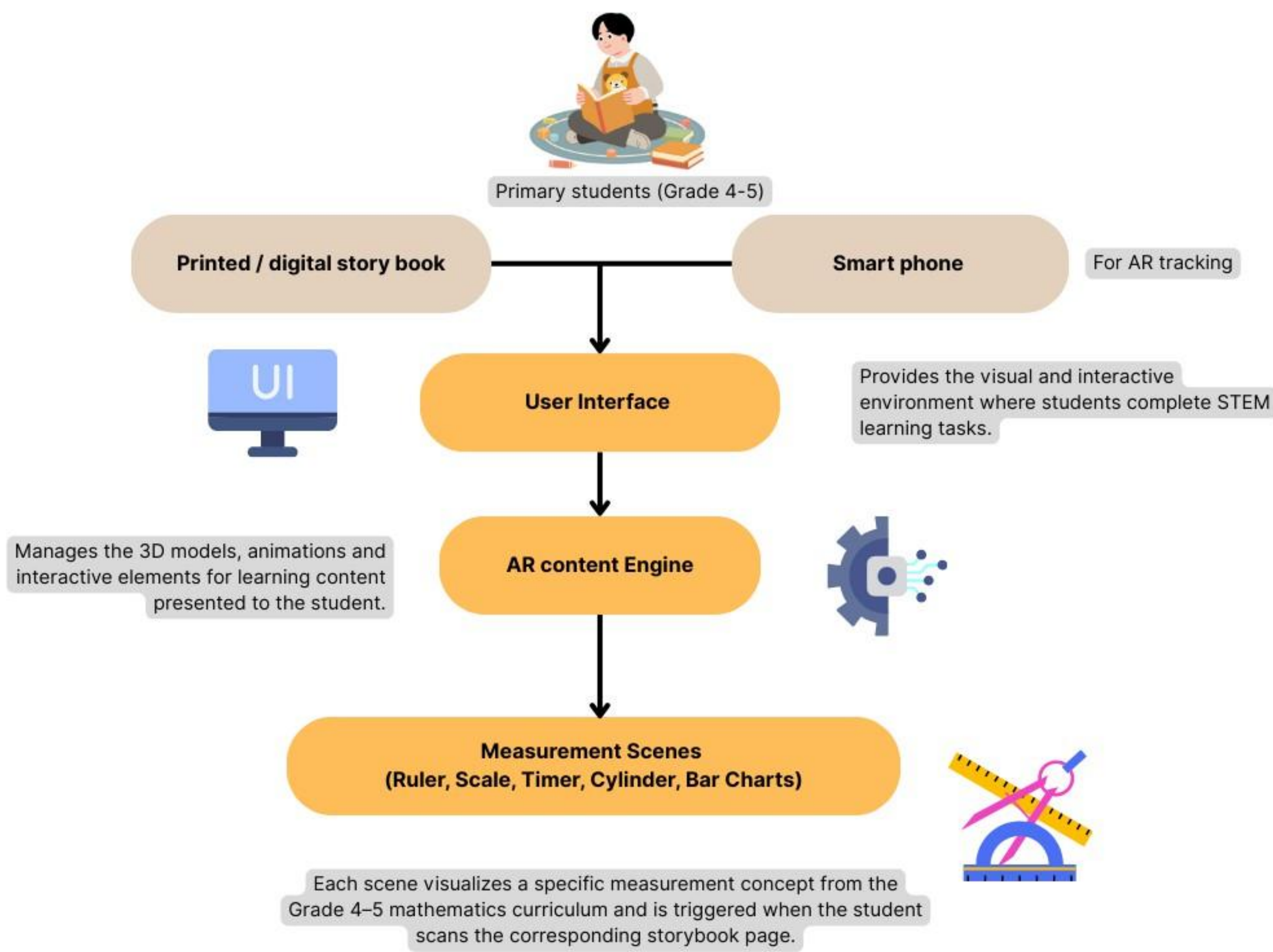
1. The student scans a storybook page using the mobile device.
2. The AR engine recognizes the page image as an Image Target.
3. The system triggers the corresponding AR scene associated with that page.
4. Interactive 3D measurement tools are displayed in the AR environment.
5. The student performs learning activities by interacting with the AR objects.
6. The system provides visual feedback and records interaction results during evaluation sessions.

The proposed architecture is designed to be scalable so that additional storybooks and learning modules can be integrated in the future. New AR scenes can be added by creating additional Image Targets and linking them to new narrative stages within the application.

Security and privacy considerations are also important, particularly because the system involves primary school students. The application does not collect personal identifying information, and any data gathered during evaluation sessions is anonymized. Additionally, the system operates locally on the device, reducing dependency on internet connectivity and ensuring safe classroom usage.

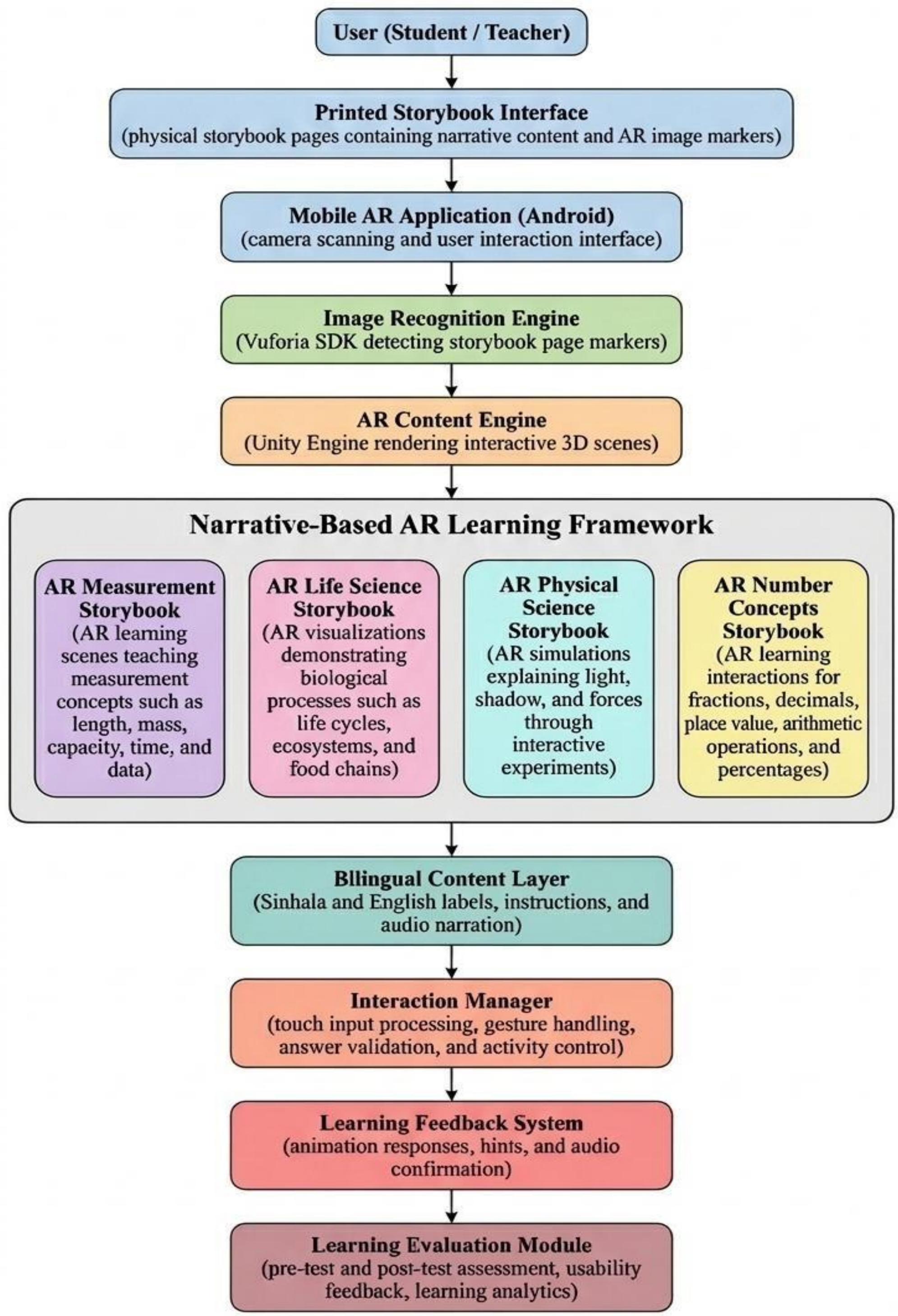
4.2 System Architecture Diagram (Component Wise)

Figure 1



4.3 Overall System Diagram (All Components)

Figure 2: Overall System Diagram



5. USER REQUIREMENTS

5.1 Stakeholder Identification

1. **Primary school students (Grade 4-5)**
   Main users of the system who interact with the AR storybook to learn measurement concepts.
2. **Teachers**
   Facilitate classroom use of the AR learning tool and guide students during learning activities.
3. **Parents**
   Parents can observe their children's learning progress and engagement while using the system at home.
4. **Researchers / Developers**
   Design, develop, and maintain the AR application and analyze the research results.

5.2 Functional Requirements

Table 2: Functional Requirements

Functional Requirement	Explanation
FR1	The system shall allow students to scan printed storybook pages using a mobile device camera.
FR2	The system shall detect image targets from the storybook using marker-based AR technology and shall display corresponding AR scenes when a valid marker is detected.
FR3	The system shall render interactive 3D measurement tools such as rulers, weighing scales, stopwatches, and measuring cylinders.
FR4	The system shall allow students to observe and interpret measurement values displayed in the AR environment.
FR5	The system shall provide immediate visual feedback when students correctly identify measurement values.
FR6	The system shall present instructions and labels in both Sinhala and English to support bilingual learning.

5.3 Non-Functional Requirements

Table 3: Non-Functional Requirements

Non-Functional Requirement	Explanation
NFR1 (Performance)	The system should respond quickly to user interactions.
NFR2 (Usability)	The interface should be simple and suitable for young learners (grade 4-5)
NFR3 (Reliability)	System should work without crashes during learning sessions.
NFR4 (Security and Privacy)	No personal data should be collected and all data must be anonymous.
NFR5 (Compatibility)	The application should run on common mobile devices used in schools without requiring specialized hardware.
NFR6 (Accessibility)	The system should support bilingual content (Sinhala and English) to accommodate different language backgrounds.
NFR7 (Scalability)	The system architecture should allow additional AR storybooks or learning modules to be added in the future.

6. COMMERCIALIZATION PLAN

The primary target market for the proposed AR learning system is primary schools and educational institutions in Sri Lanka, particularly those teaching Grade 4 and Grade 5 mathematics and science. Schools that aim to integrate digital learning tools into classroom teaching represent the main institutional customers.

The system can also target private tuition centers and educational technology providers that support students preparing for the Grade 5 Scholarship Examination. Additionally, parents of primary school students who seek interactive educational tools for home learning may also be potential customers.

The primary user persona is students aged 9 and 10 years who are learning measurement concepts in mathematics. Secondary users include teachers who guide classroom learning and integrate the AR storybook into lessons.

The proposed solution provides a unique combination of storytelling and augmented reality to make abstract measurement concepts easier for primary school students to understand. Unlike traditional textbooks that rely on static diagrams, the AR storybook allows students to interact with virtual measuring instruments such as rulers, weighing scales, stopwatches, and measuring cylinders.

The key value propositions include:

- Improved conceptual understanding through interactive 3D visualization.
- Higher student engagement through narrative-driven learning.
- Curriculum alignment with the Sri Lankan Grade 4–5 mathematics syllabus.
- Bilingual learning support in Sinhala and English.
- Low-cost accessibility, as the system runs on standard smartphones or tablets.

The commercialization model can follow a hybrid educational product strategy combining physical and digital components.

Possible revenue streams include:

- Sale of printed AR storybooks to schools and students.
- Mobile application access, provided either as a free download with limited content or a premium version with full learning modules.
- Institutional licensing, where schools purchase a license to use the AR learning system in classrooms.
- Partnerships with educational publishers to distribute AR-enabled textbooks.

The main cost categories involved in developing and deploying the system include:

- Content development costs (storybook illustrations, educational content design).
- Software development costs (AR application development using Unity and Vuforia).
- 3D asset creation costs (modeling and animation of measurement tools).
- Printing costs for the physical storybooks.
- Testing and evaluation costs during classroom experiments.

Once the system is developed, operational costs are relatively low since the application runs locally on mobile devices and does not require extensive cloud infrastructure.

The pricing strategy should aim to make the product affordable for schools and families while maintaining sustainability for developers.

Possible pricing options include:

- Low-cost printed AR storybooks sold individually to students.
- School package bundles that include multiple books and classroom licenses.
- Freemium mobile application model, where basic AR functionality is free and advanced learning modules are available through a premium upgrade.

This approach ensures accessibility while allowing long-term product sustainability.

The proposed system offers several advantages over existing educational products:

- Integration of AR with narrative storytelling for interactive learning.
- Curriculum-specific content designed for the Sri Lankan primary education system.
- Culturally relevant contexts that students can easily relate to.
- Bilingual learning support, improving accessibility for different language groups.
- A mobile-based learning platform that requires no specialized equipment.

These factors differentiate the system from conventional textbooks and generic educational apps.

And for the intellectual property of the system may include:

- Copyright protection for the storybook content, illustrations, and AR visual assets.
- Software ownership of the AR mobile application developed using Unity and Vuforia.
- Potential trademark registration for the product name and branding of the AR storybook series.
- Protecting these assets ensures that the developed system can be safely commercialized and expanded into additional AR educational products in the future.

7. BUDGET AND JUSTIFICATION

Table 4: Budget and Justification

Item	Description	Justification	Estimated Cost (LKR)
Laptop / Development machine	Required for Unity development and AR simulation	Existing personal laptop will be used for development.	-
Smartphone (Testing device)	Used to test the AR application with marker detection	Personal mobile device will be used for testing and demonstration.	-
Development Software	Unity Game Engine and Vuforia AR SDK	Free educational versions are available	-
3D Asset Creation	Modeling of measurement tools	Some assets may be purchased from online asset stores or created with design tools.	5,000
Storybook Design	Illustration and layout design for the AR storybook pages	Illustration and layout design for the AR storybook pages	7,000
Printing of Storybook Prototype	Printing the marker-based narrative learning book	Physical copies required for marker-based AR interaction and user testing.	7,500
Data Storage & Backup	External drive / cloud storage for project files	For the application development and store user data	8,000
Miscellaneous Development Costs	Internet, small printing, testing materials	Covers additional small development or testing costs.	5,000
Total Estimated Budget			**32,500**

Overall, the project is designed to be cost-efficient, utilizing existing hardware resources and freely available development tools while focusing expenditure on content development and testing activities.

8. WORK BREAKDOWN STRUCTURE (WBS)

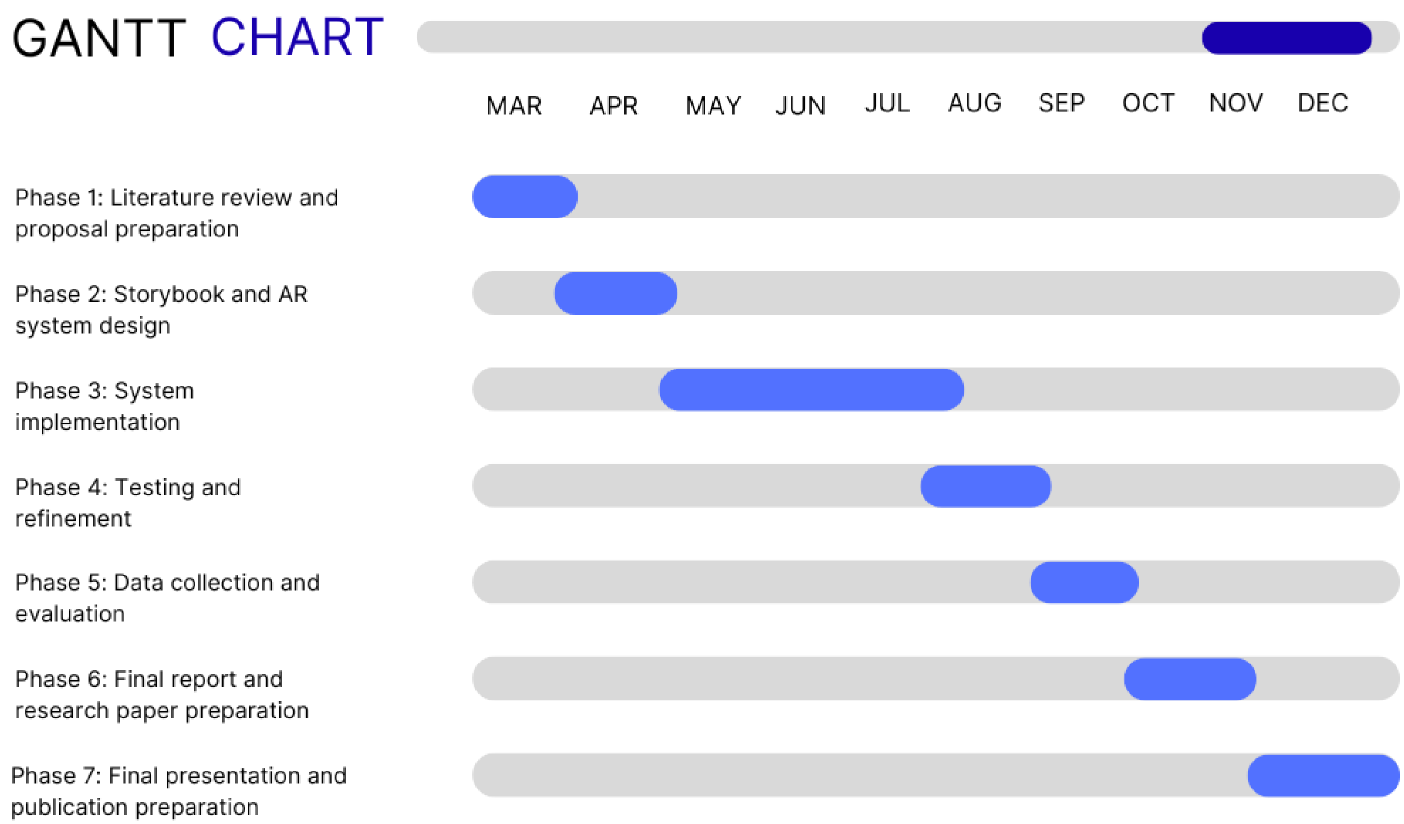
Table 5: Work Breakdown Structure

Phase	Activity
Phase 1	Literature review and proposal preparation Tasks include: • Reviewing recent literature on AR-based education and interactive learning systems. • Studying existing AR learning applications used for primary education. • Identifying limitations in current digital learning tools related to measurement concepts. • Defining research objectives, methodology, and system concept for the AR measurement storybook.
Phase 2	Storybook and AR system design Key tasks include: • Designing the narrative concept for the AR measurement storybook. • Identifying measurement learning scenes. • Designing storybook pages and AR markers for image recognition. • Defining the interaction flow between the storybook and AR scenes.
Phase 3	AR System implementation Main tasks include: • Setting up the development environment. • Creating 3D models and visual assets for measurement tools. • Implementing marker-based AR scenes linked to storybook pages. • Integrating interactive elements and bilingual labels for learning support.
Phase 4	Testing and refinement Key tasks include: • Testing AR marker recognition and scene rendering accuracy. • Evaluating system responsiveness and interaction flow. • Identifying and fixing implementation errors. • Optimizing system performance for smooth execution on mobile devices.
Phase 5	Data collection and evaluation Tasks include: • Collecting anonymous learner interaction data during system usage. • Collecting feedback on engagement and concept understanding. • Recording observations related to usability and learning interaction • Analyzing results to evaluate the effectiveness of the AR learning approach.

Phase 6	Final report and research paper preparation Tasks include: • Writing the final research report. • Preparing the research paper. • Developing the project website. • Preparing the final presentation and viva materials.
Phase 7	Final presentation and publication preparation Tasks include: • Presenting the completed system during the final project presentation. • Submitting the final research paper for publication. • Providing research paper publication evidence and completing final documentation requirements.

9. GANTT CHART

Figure 3: Gantt Chart



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